

Overview of AUS-M

AUS-M is essentially an outgrowth of the 1990s version of the Treasury Macroeconomic (TRYM) model. TRYM in turn developed out of the AMPS model developed by the EPAC Secretariat in the 1980s, as did the Murphy model.¹ AMPS, TRYM and the Murphy Model were all models in the Keynes Klein (K-K) tradition with sticky prices and wages and output determined by demand in the short-term. Where they differed from early applied K-K models such as the 1971 Treasury NIF model, was in their response to the Lucas and Sims critiques of the theoretical and econometric underpinnings of these models. Like other models of the 1990s such as the US FRB MPS model, the IMF's Multi-Mod, the Bank of England's core macro model, and Warwick McKibbin's MSG model, they introduced explicit optimisation to the supply side of the model, took greater care in the treatment of expectations, had a derivable steady state solution, and used econometric techniques that took greater note of orders of integration and issues of dynamic consistency (the consistency of short-run dynamics with the long-run steady state). AMPS, TRYM and Murphy treated the financial sector as rational and forward looking, but household and business sectors as adaptive and backward-looking, (although in TRYM there were some elements of forward looking behaviour – for example Q ratios and wealth variables would jump with changes in underlying productivity growth).

Given that the properties of the TRYM model are well documented and were the subject of extensive scrutiny and validation, it is probably useful to focus on the key points of difference between the two models. The main difference is the way in which demand and supply linkages are treated in AUS-M compared with the earlier model. The first section below provides a broad overview of how the industry and commodity detail has been introduced in AUS-M, followed by a short summary of some of the other developments in the model.

Extensive documentation of the TRYM model including background papers is available at:

<http://archive.treasury.gov.au/contentitem.asp?NavId=016&ContentID=238>

Supply and Demand in AUS-M

AUS-M retains many of the overarching features of TRYM, and has similar macroeconomic properties as a result. (Those in TRYM were extensively tested, subject to academic and peer review, independently replicated by outside groups, and reconciled with the results from single equation, VAR and other models.) A fiscal or monetary policy change in AUS-M for example gives similar results to the original TRYM model. The main difference is in the wealth of industry, commodity and other detail provided in the results.

¹ See Murphy et al 1986 on the AMPS model.

Table 2 below shows the supply and demand variables in AUS-M. The way the detail is incorporated, and supply and demand are linked is by utilising information from the input-output tables. To do so with a minimum of complexity the input-output tables are first reduced down to a manageable size by inverting the tables to produce a mapping of supply to demand.² This mapping is then used to form up a demand system for the supply side variables (industry output and imports – see Figure 3). A framework similar to the Almost Ideal Demand System³ is used to model industry and import volumes with each individual supply component depending on: (1) the input output weighted demand term (2) relative prices, (3) the change in total demand and (4) a stochastic time trend to capture effects from changing technology and tastes. More detail on the industry and import demand system is provided in Appendix A.

² This table is known as the Primary Input Content of Final Demand. If: I is the identity matrix; A the matrix of industry to industry interactions; and D the matrix of industry to final demand coefficients, then the table is given by $[I-A]^{-1}D$ (i.e. the Leontief inverse times the final demand component of the supply-use tables). As the ABS national accounts produce chain-linked, constant-price, time-series estimates of industry value added we are mainly interested in the allocation of value added to final demands (and vice versa) thus avoiding much of the detail of a full blown CGE model. To build an econometric time-series model consistent with the historical data we need to focus on the available time series data. There is no available time series data for the thousands and thousands of industry to industry interactions. Hence we solve out for the industry to industry interactions and model them in a reduced form way by forming up a demand system for individual industry output, that depend on relative prices and the allocation out of demand items. That is, rather than impose a set of industry to industry substitution effects by calibrating an extremely detailed set of equations based on imputed data, we let those effects be determined by the responses to relative prices evident in the historical time series data.

³ Deaton and Muelbauer, 1992.

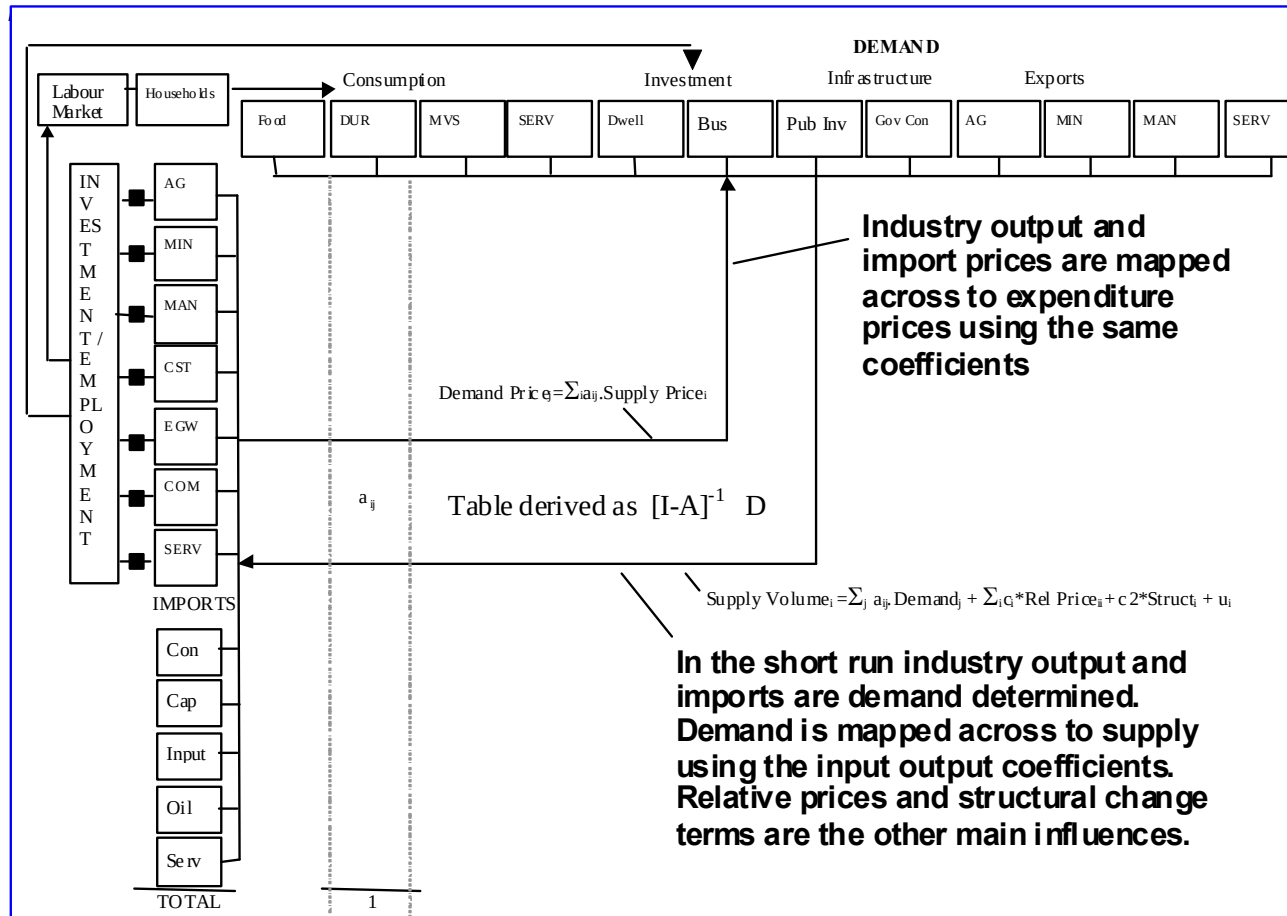
1 Supply and demand variables in AUS-M

<i>Supply variable</i>	<i>Inemonic</i>	<i>Price</i>	<i>Demand Variable</i>	<i>Inemonic</i>	<i>Price</i>
<i>Domestic Supply:</i>			<i>Domestic Demand:</i>		
Agriculture	GAG	PGAG	Food	CFD	PCFD
Mining	GMIN	PGMIN	Cigarettes and Alcohol	CCAL	PCCAL
Manufacturing	GMAN	PGMAN	Durables and other Goods	CDOG	PCDOG
Construction	GCST	PGCST	Services from Durables	CSDOG	
Retail Trade	GRET	PGRET	Motor Vehicles	CMV	PCMV
Wholesale Trade	GWST	PGWST	Services from Motor Vehicles	CSMV	
Transport Services	GTRN	PGTRN	Operation of motor vehicles and trans	COMT	PCOMT
Distributional Services	GSD	PGSD	Electricity Gas and Water	CEGW	PCEGW
Accomodation & Food Services	GACC	PGACC	Communications	CCOM	PCCOM
Arts & Recreational Services	GREC	PGREC	Health and Education	CHUC	PCHUC
Other Services	GOTS	PGOTS	Other Services	CSRV	PCSRV
Consumer Services	GSC	PGSC	Adding up discrepancy (consumption)	DISCON	
Real Estate and Rental Services	GRES	PGRES	Non-rental consumption	CNR	PCNR
Professional and Technical Services	GPRO	PGPRO	Rental Consumption	CRE	PCRE
Administrative and Support Services	GADS	PGADS	Total Consumption	CON	PCON
Property and Business Services	GPBS	PGPBS	Dwelling investment	IDW	PIDW
Finance and Insurance	GFIN	PGFIN	Non-dwelling construction	IOB	PIOB
Communications	GCOM	PGCOM	Machinery and equipment	IFE	PIFE
Electricity Gas and Water	GEGW	PGEGW	Second hand asset sales	IBS	
Health Care & Social Assisstance	GHLC	PGHLC	Private Business Investment	IBP	PIBP
Education and Training	GED	PGED	Public Enterprise Investment	IGE	PIGE
Public Administration and Safety	GPAD	PGPAD	General Government Investment	IGG	PIGG
Human Capital Services	GHUC	PGHUC	General Government Consumption	CONG	PCONG
Dwelling services (owned by households)	GDW	PGDW	Mining stock building	SMIN	
Dwelling services (total dwellings)	GDWT	PGDWT	Manufacturing stock-building	SMAN	
Taxes less subsidies on products	GITS	PGITS	Wholesale and retail stock-building	SSD	
Adding up discrepancy (production)	DISADP		Other industries stock-building	SOTH	
Gross Domestic Product Production	GDPP	PGDPP	Non-farm stock-building	SNN	PSNN
Statistical discepancy (GDPA-GDPP)	DISP		Farm stock-building	SFM	
<i>Imported Supply</i>			General Government stock-building	SBG	
Imports of Consumer Goods	MCON	PMCON	Additive Discrepancy (Expenditure)	DISEAD	
Imports of Capital Goods	MCAP	PMCAP	Gross Domestic Product (Expenditure)	GDPE	PGDPE
Imports of Intermediate Goods	MINT	PMINT	Statistical Discrepancy (GDPA-GDPE)	DISE	
Imports of Services	MSRV	PMSRV	<i>External Demand:</i>		
Imports of Petroleum Products	MPE	PMPE	Agricultural Exports	XAG	PXAG
Imports of Information Technology	MGIT	PMGIT	Mining Exports	XMIN	PXMIN
Imports of Civilian Aircraft	MCA	PMCA	Manufacturing Exports	XMAN	PXMAN
Imports of Non-Monetary Gold	MNMG	PMNMG	Service Exports	XSRV	PXSRV
Adding up discrepancy (imports)	DISMGS		Adding up discrepancy (exports)	DISXGS	
Imports of Goods and Services (Total)	MGS	PMGS	Exports of Goods and Services Total	XGS	PXGS

Notes: The table shows the main variables. Note that business investment is modelled on an industry basis so that there are investment volume and price variables plus variables for sub-components by asset class for each individual industry. These are aggregated up to produce the investment variables shown.

Data source: Outlook Economics, AUS-M Model Database.

2 Stylised representation of the input-output core of the AUS-M model



The other key elements of AUS-M in addition to those in TRYM are:

- *The consumption demand system* – to disaggregate household consumption into its component parts. Like the demand system for imports and industry outputs this is based on an almost ideal demand system style framework where each individual component is estimated on the basis of relative prices, the change in total consumption to pick up cyclical and income effects, and stochastic trends to capture changing tastes and technology over time.
- *The estimated system of equations for expenditure deflators*. This system maps across supply side deflators to the demand-side expenditure items. Each demand side deflator (P.) has an input-output weighted supply side equivalent (PP.). Indirect taxes are also allocated out using the input output coefficients leading to implicit tax variables for each demand component (RT.). Expenditure deflators (P.) respond directly to supply prices (PP.) indirect taxes (RT.) and the GST.
- *Disaggregated production function equations* – for employment, investment and output prices. The parameters of the production function for each industry are estimated indirectly by jointly estimating equations for employment, investment and output prices. (The long run of each equation depends on the marginal product condition for profit maximization/cost minimisation and hence the equations contain common parameters.) Stochastic trends capture changes in underlying labour and capital productivity. Each industry has a derived measure of potential output and hence capacity utilization which can be compared with measures from the business surveys.
- *Disaggregated trade equations* – for import and export prices and export volumes. These involved forming up a range of new trade weighted exchange rates, world prices, and weighted trading partner demand terms.
- *Disaggregated equations for inventories*. Stock-building is broken down into its component parts (mining, manufacturing, distributional services, agriculture). Each component has a separate measure of sales which is related back to the demand terms for the individual sectors. In the long run inventories return to a level consistent with the equilibrium stocks to sales ratio for each industry.

The model also includes much more detail in a number of other areas with a particular focus on markets that are characterised by search, partial equilibrium and stock-adjustment dynamics. For example:

- The labour market framework includes unfilled job vacancies and a detailed cohort model that forecasts the duration structure of unemployment. The cohort model contains equations for exit rates (from unemployment) which capture turnover effects. As the economy goes into recession, employment turnover falls, exit rates from unemployment fall and long term unemployment increases. As the long term unemployed are less active in job search, this reduces the overall search effectiveness of the unemployed, pushing up the vacancy rate for any given level

of unemployment. A decrease in search effectiveness also reduces the impact of any given level of unemployment on wage inflation (i.e. raises the NAWRU).

- Movements in vacancies also impact on labour demand. When the economy is close to or above full employment, increases in labour demand will tend to manifest themselves more in terms of an increase in unfilled vacancies than an increase in employment (reducing the volatility of employment). Employment equations are estimated on an industry basis, and employment demand for each industry is adjusted for unfilled vacancies and for average hours worked for the particular industry.
- Employment by industry is broken down into its broad occupational components, allowing quality adjusted labour inputs to be calculated for each industry. (This is also useful for explaining aggregate wage behaviour, for part of the movement of wages over the cycle is due to the changing composition of employment as the unskilled are the first to lose their jobs in a downturn.)
- The labour force participation equation is broken down by age and sex allowing for more detailed examination of both underlying trends and cyclical encouraged/discouraged worker effects.
- The dwelling sector includes an equation for rental vacancies, in turn dependent on demand versus short-term supply. Lower vacancies lead to higher rents. Higher rents lead to higher rental returns, a higher dwelling Q ratio, and a higher level of investment (higher supply). Higher rental prices also lead to a substitution away from the consumption of dwelling services (lower demand). A higher dwelling Q ratio is associated with higher established house prices, leading to higher consumption (via the wealth effect).
- Explicit equations have been developed for non-dwelling construction investment in Finance and Insurance, Property and Business Services and Consumer Services to capture some of the movement in commercial property investment. This involves equations for both vacancies, and investment. Vacancies are largely dependent on industry employment growth relative to the capital stock, feeding back to the incentive to invest, (but with long time-to-build lags). Adjustments are made to the aggregate industry investment equations to account for the consequent cyclical ups and downs. (For example the office building boom bust cycle is mainly reflected in investment in Property and Business Services, and Finance and Insurance.)
- Government budget identities are much more detailed, reflecting the greater detail available in the model to establish various tax bases, and the fact that the model has a complete representation on the income side adding up to GDPI.

Extensive use is made of joint estimation and systems estimation in constructing the model. In one sense the model can be viewed as a kind of Sudoku, utilizing constraints across a number of dimensions to reconcile estimates and fill in a complete picture of how the economy works. Anything on the expenditure side has

implications for the production side and the income side, and vice versa, both in terms of volumes and prices. The model forecasts converge on a steady state growth path that essentially stems from a small scale CGE growth model (the steady state version of the model).

Caveat

Like all models, AUS-M is only an approximation of reality and there are a number of limitations which need to be kept in mind. There are many specification choices, places where data is insufficient, or where more detail is required. Human behaviour is not constant and is subject to learning, rationality is bounded, and economic structures and institutions are constantly changing. Economic analysis that involves feedback is complex. Consequently, the model is not meant to be used just in a mechanical way, and results should be interpreted with care. It will usually require some work to make appropriate adjustments to the model and to consider the linkages and interpret the results. In that sense it is not meant to be a substitute for hard work, but rather and ideally a complement to thoughtful and more detailed analysis.

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<http://archive.treasury.gov.au/contentitem.asp?ContentID=238&NavID=>